Illustration

A family is selected at random from the set of all families in a town with one twin pair. The sample space is

$$S = \{(\text{boy, boy}), (\text{boy, girl}), (\text{girl, boy}), (\text{girl, girl})\}.$$

Define the events

A: the randomly selected family has two boys, and

B: the randomly selected family has a boy.

Let us assume that all the families with one twin pair are equally likely. Since

$$A = \{(\text{boy, boy})\},$$

$$B = \{(\text{boy, boy}), (\text{boy, girl}), (\text{girl, boy})\},$$

$$A \cap B = A = \{(\text{boy, boy})\}.$$

Applying the classical definition of probability, it can be calculated that

$$P(B) = \frac{3}{4} \text{ and } P(A \cap B) = \frac{1}{4}.$$

Suppose that the randomly selected family has a boy. Then, the probability that the other child in the pair is a girl can be calculated using conditional probability as

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{1/4}{3/4} = 1/3.$$

Example 8.14

A number is selected randomly from 11 through 19. Consider the events

$A = \{ 11,14, 16, 18, 19 \}$ $B = \{ 12, 14, 18, 19 \}$ $C = \{ 13, 15, 18, 19 \}$.

Find (i) P(A/B) (ii) P(A/C) (iii) P(B/C) (iv) P (B/A)

Solution:

given A $= \{ 11,14, 16, 18, 19 \}$ $B = \{ 12, 14, 18, 19 \}$ $C = \{ 13, 15, 18, 19 \}$.

$$A \cap B = \{ 14, 18, 19 \} \quad A \cap C = \{ 18, 19 \} = B \cap C$$

$$P(A) = \frac{5}{9} \qquad P(B) = \frac{4}{9} = P(C)$$

$$P(A \cap B) = \frac{3}{9} \qquad P(A \cap C) = \frac{2}{9} = P(B \cap C)$$